COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Tutorial Sheet: Unit 2 – Sessions 1, 2 & 3 (Sections A & B)
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Section A – Very Short Answers

• 1. Four main functional units: Input Unit, Output Unit, Memory Unit, and CPU (Arithmetic Logic Unit + Control Unit).

• 2. Control Unit: coordinates and controls all other units — it fetches and decodes instructions and issues timing/control signals that execute them.

• 3. Bus: a shared communication pathway (set of lines) that transfers data, addresses and control signals between CPU, memory and I/O devices.

• 4. ALU: the Arithmetic Logic Unit — the part of the CPU that performs arithmetic (add, subtract) and logic (AND, OR, compare) operations.

• 5. Address bus is used to transfer memory addresses.

Section B – Short Answers

• 1. Data Bus carries actual data between units (bidirectional); Address Bus carries the location/address of the data (unidirectional, CPU → memory); Control Bus carries command and timing signals such as Read, Write and Interrupt.

• 2. Program Counter: holds the address of the next instruction; during fetch its value goes to the MAR, then it auto-increments (or is set by a branch) so instructions execute in sequence.

• 3. Steps for Load R2, LOC: (i) PC → MAR, instruction fetched to IR, PC incremented; (ii) instruction decoded; (iii) address field LOC → MAR, memory read; (iv) data → MDR; (v) MDR → R2.

• 4. Limitations of single bus: only one transfer at a time (serialized), bus contention slows the system, performance limited by bus speed, a bus fault can halt the whole system.

• 5. Advantages: simple/low cost, easy to connect devices. Disadvantages: slower due to sharing, not suitable for high-performance systems with many transfers.